Non-terms

All literals are non-terms. For any non-terms x and y, $xy, (x)y, x(y), (xy)$, and (x) are all non-terms. The termified form of an expression $x_1x_2...x_{n-1}x_n$ where all x_i are non-terms, is (mn) where m is the termified form of $x_1x_2...x_{n-1}$ and n is the termified form of x_n. The base cases for the termified forms are as follows: the termified form of x is x for all literals, and, for all non-terms x, the termified form of (x) is n where n is the termified form of x.

We can prove that, for all non-terms x, the termified form of x is a term, by complete induction. The base case is just when x is a constant or a variable. Thus the termified form of x is just x itself, and because it is a constant or a variable, it is also a term. Now suppose the result holds for all numbers less than n. Consider any non-term x of length n. It is either of the form (y) where y is a non-term, or of the form $x_1x_2...x_{m-1}x_m$ where all x_i are non-terms. In the first case, the termified form of x is a where a is the termified form of y. But by the induction assumption, the termified form of y is a term. Hence the termified form of x is a term. In the second case, the termified form of x is (ab) where a is the termified form of $x_1x_2...x_{n-1}$ and b is the termified form of x_n. But by the induction assumption, a and b are terms, meaning so is (ab), and thus the termified form of x is a term, therefore completing the proof.

It should also be noted that for all terms t, the termified form of t is just t. The base case when t is a constant or a variable is obvious. Now suppose the result holds for all terms of length less than n. Now consider some non-term of lenth n. By definition, it is of the form (t_1t_2) where t_1 and t_2 are both terms. Thus the termified form of t is (xy) where x and y are the termified forms of t_1 and t_2, respectively. But by the induction assumption, $x = t_1$ and $y = t_2$. Thus $(xy) = t$.